

ITS and LSU nrDNA analyses

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Internal Transcribed Spacer (ITS) region of the nuclear ribosomal gene, including the 5.8S subunit (ITS nrDNA), the current barcoding of Fungi [1], and Large Subunit region of nuclear ribosomal DNA (LSU nrDNA) were the loci selected for molecular analyses. After DNA extraction, the ITS regions were amplified and sequenced using the primer pair ITS1F/ITS4 [2,3] for all samples. Part of the LSU was amplified and sequenced using either the primer pair LR0R/LR5 [4] for UFRN-Fungos, INPA-Fungos and CEPEC samples or the primer pair CTB6-TW14 [3] for the other samples. When amplifications were not obtained, nested PCR of ITS and semi-nested PCR of LSU were performed: first amplification with primers ITS1F/ITS4B [2] and LR0R/LR7 [4], and a second amplification with ITS5/ITS4 [3], and LR0R/LR5 or LR0R/LR4 [4]. Amplifications were done using illustra™ PureTaq™ Ready-To-Go™ PCR Beads (Healthcare, Buckinghamshire, UK), adding 1 µl at 10 µM of each primer and 23 µl (1.5–5.0 ng/µl) of isolated DNA, and cycling condition indicated in Martín and Winka [5]. Verification of amplification products was done on 1% agarose gels (UltraPure™ Invitrogen). Purification was done with ExoSAP-IT® (USB Corporation, OH, USA) and the samples were sequenced bidirectionally in MacroGen Inc. (Seoul, South Korea). Consensus sequences were obtained using Sequencher 5.2.4 (Gene Codes Corp., USA). Preliminary identifications were performed through megablast searches comparing the newly-generated sequences with those in GenBank [6]. Sequences obtained in this study were submitted to Genbank under the accession numbers indicated in Table 1.

The newly-generated ITS and LSU sequences, and homologous sequences retrieved from EMBL/GenBank/DDBJ databases. The molecular dataset of unambiguously aligned ITS sequences was submitted to three types of analyses: maximum parsimony (MP), maximum likelihood (ML), and Bayesian inference. Concatenate ITS/LSU analyses were also carried out through each method, but concatenated datasets included only those specimens which have sequences for both ITS and LSU nrDNA.

The software PAUP (Phylogenetic Analysis Using Parsimony) version 4.0a (build 159) for 32-bit Microsoft Windows [7] was used for MP analysis, in which minimum length Fitch trees were calculated by heuristic search with the Tree Bisection Reconnection (TBR) algorithm for branch-swapping, collapsing branches if maximum length is zero, and Multitrees option in effect. The trees were obtained by 2 random stepwise addition, and bootstrap of 10,000 replicates (MPbs); gaps were treated as a new state. The consistency index, CI [8], retention index, RI [9], rescaled consistency index, RC [8] and homoplasy index, HI [10], were obtained for the summarized 50% majority rule consensus tree.

ML analyses were performed by Randomized Accelerated Maximum Likelihood (RAxML HPC-2, [11] implemented in the Cyberinfrastructure for Phylogenetic Research web portal (CIPRES Science Gateway, <http://www.phylo.org/>). The analyses were performed with Lewis ascertainment bias correction and 1,000 bootstrap replicates (MLbs) under the default GTRCAT model. A 50% majority rule consensus tree was calculated from the resulting trees dataset.

Bayesian analyses were conducted with BEAST v2.4.7 software [12] along with the utility software BEAUti v2, LogCombiner v2.4.7, and TreeAnnotator v2.4.3. Both analyses from ITS alone and concatenated ITS/LSU consisted of two different runs with 50 million MCMC generations under the Yule model of speciation, sampled every

5,000 generations. Trees were sampled every 5,000 generations, and evolutionary methods were chosen by JmodelTest 2 software [13] for each marker. After checking for convergence between runs on Tracer v1.7 [14] a portion of trees was discarded as burn-in until the stationary phase. The remaining sampled trees were used to calculate Maximum Clade Credibility (MCC) trees.

Trees resulted from MP, ML and Bayesian analyses were visualized and edited with FigTree (<http://tree.bio.ed.ac.uk/software/figtree/>) and CorelDraw® Graphics Suite X8.

Terminal nodes which were not supported in all three analyses were reanalyzed using an ITS nrDNA alignment containing only samples which, according morphology, needed to be delimited based on sequences. In this alignment, sequences were cut to avoid short nucleotide coverage at the beginning and at the end, reducing uncertainty in the data. This alignment was submitted to Bayesian analysis in BEAST 2 software, following the same methodology described above. The resulting MCC tree was used as input for creating a Topo-Phylogeny representation [15], and a Phylogenetic Network with SplitsTree4 [16] in order to better visualize species assignment. Final artwork was done in CorelDraw® Graphics Suite X8.

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